

Android Calculator Project Report

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Project Title

Android Calculator Application

Objective

To develop a simple calculator application using Android Studio that performs basic arithmetic operations with a user-friendly interface.

Tools Used

- Android Studio
- Java/Kotlin
- XML
- Android SDK
- Android Emulator/Physical Device

Features

- Addition (+)
- Subtraction (−)
- Multiplication (×)
- Division (÷)
- Percentage (%)
- Positive/Negative (±)
- Delete (□)
- Decimal Point (.)
- Displays calculation expression and result

Implementation

1. Created a new Android Studio project.
2. Designed the calculator interface using XML.
3. Added buttons and display screens.
4. Programmed button click events using Java/Kotlin.
5. Tested the application on an Android emulator.

Testing Results

Test	Expected	Result	Status
$5 + 5$	10	10	Passed
$20 - 8$	12	12	Passed
9×9	81	81	Passed
$40 \div 5$	8	8	Passed
98×65	6370	6370	Passed

Errors Encountered

- Gradle build failed.
- Emulator failed to start.
- App crashed due to coding errors.
- Buttons were not responding.
- XML syntax errors.
- SDK or Gradle synchronization issues.
- Division by zero caused calculation errors.

Solutions

- Synced and rebuilt the project.
- Restarted Android Studio and emulator.
- Corrected XML and Java/Kotlin code.
- Checked Logcat for debugging.
- Installed the correct Android SDK.
- Added input validation for division by zero.

Precautions

- Save the project regularly.
- Test the app after every change.
- Keep Android Studio and SDK updated.
- Use correct XML IDs and syntax.
- Validate user inputs.
- Check Logcat for errors.
- Back up the project files.

Conclusion

The Android Calculator application was successfully developed and tested. It performs basic arithmetic operations accurately and provides a clean, modern interface. The project improved practical skills in Android development, user interface design, debugging, and testing.